

A Complete, From–Scratch Derivation of the Thermodynamic Neuron

From a single qubit to the autonomous NOT gate and the general linearly–separable neuron

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(pedagogical companion to Lipka–Bartosik, Perarnau–Llobet & Brunner, *Sci. Adv.* **10**, eadm8792, 2024)

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Abstract

This note rebuilds the “thermodynamic neuron” of Ref. [1] from the ground up, at the level of detail a student needs in order to reproduce every equation by hand and in code. We start from a single qubit in a heat bath and derive, step by step: (i) the thermal (Gibbs) state and the Fermi–Dirac/sigmoid occupation; (ii) the open–system master equation, showing explicitly *where the bath Hamiltonian goes* and *why there is no surviving qubit–bath interaction Hamiltonian in the final equations* — the single most common point of confusion; (iii) the reset (collisional) thermalization model and its heat current; (iv) the three–qubit machine, the *virtual qubit*, and the *virtual temperature*; (v) the full interaction Hamiltonian of the collector and why it must be a three–body, energy–conserving swap; (vi) why the collector alone is only a linear amplifier and hence *why the modulator is needed*; (vii) the two–time–scale dynamics, the steady–state transfer characteristic, and the emergence of the sigmoid; (viii) logic encoding, readout error, and the error–dissipation trade–off; and (ix) the generalization to any linearly–separable Boolean function and its exact correspondence with the perceptron. Every approximation is called out in an orange box (*where, why, when*), and every result is accompanied by a blue box explaining *how to implement it* in QuTiP. Equation numbers of the source paper are quoted throughout as “[paper Eq. (n)]”.

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Reading guide

The derivation is organized as a ladder. Each rung is self-contained and ends with the object the next rung needs.

Part What is built

- I One qubit $\rightarrow$ open-system master equation $\rightarrow$ reset model $\rightarrow$ multi-qubit machine.
 - II Two qubits $\rightarrow$ virtual qubit $\rightarrow$ virtual temperature $\rightarrow$ resonant three-body interaction.
 - III Collector + modulator $\rightarrow$ dynamics $\rightarrow$ transfer curve $\rightarrow$ sigmoid $\rightarrow$ NOT gate $\rightarrow$ error/dissipation.
 - IV n -input neuron $\rightarrow$ perceptron equivalence $\rightarrow$ design algorithm $\rightarrow$ NOR, MAJORITY, XOR networks.
-

Three questions are answered repeatedly and are flagged with coloured boxes:

- **Approximation boxes** state *which* approximation is made, *why* it is legitimate, and *when* (in which parameter regime) it holds.
- **Implementation boxes** translate each result into concrete QuTiP objects.

Part I

Physical foundations: from one qubit to a thermal machine

1 The single qubit

1.1 Hilbert space, states, Hamiltonian

A qubit is a two-level quantum system. Its Hilbert space is $\mathcal{H} = \mathbb{C}^2$, with an orthonormal basis given by the ground state $|0\rangle$ and the excited state $|1\rangle$,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \langle i|j\rangle = \delta_{ij}. \quad (1)$$

A general (possibly mixed) state is a density operator ρ : a positive semidefinite, Hermitian, unit-trace operator,

$$\rho = \rho^\dagger \succeq 0, \quad \text{Tr } \rho = 1. \quad (2)$$

The free Hamiltonian assigns energies E_0 and $E_1 > E_0$ to the two levels,

$$H = E_0 |0\rangle \langle 0| + E_1 |1\rangle \langle 1|. \quad (3)$$

Only *energy differences* are physical, so we are free to set the zero of energy at the ground state, $E_0 = 0$. Then

$$H = \epsilon |1\rangle \langle 1| = \epsilon n, \quad \epsilon := E_1 - E_0, \quad n := |1\rangle \langle 1|. \quad (4)$$

The single number ϵ — the *energy gap* — completely specifies the qubit. This is exactly the Hamiltonian $H = E_0 |0\rangle \langle 0| + E_1 |1\rangle \langle 1|$ of [paper, Sec. II A 2] with $E_0 = 0$.

Remark 1.1. It is often convenient to use Pauli operators. With $\sigma_z = |0\rangle \langle 0| - |1\rangle \langle 1|$ one has $n = \frac{1}{2}(\mathbb{I} - \sigma_z)$, so $H = \frac{\epsilon}{2}(\mathbb{I} - \sigma_z)$, which equals $-\frac{\epsilon}{2}\sigma_z$ up to the physically irrelevant constant $\frac{\epsilon}{2}\mathbb{I}$. We keep the form ϵn because the “excited-state population” n is the quantity we will read out. The raising/lowering operators are $\sigma_+ = |1\rangle \langle 0|$, $\sigma_- = |0\rangle \langle 1|$, with $\sigma_x = \sigma_+ + \sigma_-$.

1.2 The thermal (Gibbs) state and the partition function

Suppose the qubit is left in contact with a large heat bath at inverse temperature $\beta = 1/(k_B T)$ for a long time. Statistical mechanics tells us it settles into the canonical (Gibbs) state

$$\tau(\beta) := \frac{e^{-\beta H}}{Z}, \quad Z := \text{Tr } e^{-\beta H} = 1 + e^{-\beta \epsilon}. \quad (5)$$

Because H is diagonal, so is $\tau(\beta)$:

$$\tau(\beta) = \frac{1}{1 + e^{-\beta \epsilon}} |0\rangle \langle 0| + \frac{e^{-\beta \epsilon}}{1 + e^{-\beta \epsilon}} |1\rangle \langle 1|. \quad (6)$$

This is [paper Eq. (8)] for a single qubit.

1.3 Excited population = Fermi–Dirac = sigmoid

The probability that the qubit is found excited is the $|1\rangle \langle 1|$ matrix element of $\tau(\beta)$,

$$g(\beta \epsilon) := \langle 1 | \tau(\beta) | 1 \rangle = \frac{e^{-\beta \epsilon}}{1 + e^{-\beta \epsilon}} = \frac{1}{1 + e^{\beta \epsilon}}. \quad (7)$$

The last equality (multiply numerator and denominator by $e^{+\beta \epsilon}$) is [paper Eq. (2)]. Three facts about g will be used constantly:

- (a) **It is the Fermi–Dirac function**, monotonically decreasing in $\beta \epsilon$, with $g(-\infty) = 1$, $g(0) = \frac{1}{2}$, $g(+\infty) = 0$.
- (b) **It is the logistic sigmoid** $f(x) = (1 + e^x)^{-1}$ evaluated at $x = \beta \epsilon$. This single identity is the seed of the entire neuron–perceptron analogy: a thermal population *is* a sigmoid of an (inverse) temperature.
- (c) **Detailed–balance ratio.** From (6),

$$\frac{\langle 1 | \tau(\beta) | 1 \rangle}{\langle 0 | \tau(\beta) | 0 \rangle} = e^{-\beta \epsilon}. \quad (8)$$

The ratio of excited to ground population is the Boltzmann factor. This is the microscopic fingerprint of temperature and is the property every thermalization model must reproduce.

How to implement — the thermal state

In QuTiP: `H = eps * create(2)*destroy(2)` (i.e. ϵn), and the Gibbs state is `tau = (-beta*H).expm(); tau/=tau.tr()`. The excited population is `expect(num(2), tau)`, which returns $g(\beta \epsilon)$ of (7). Equivalently `g = 1/(1+np.exp(beta*eps))`.

2 Open quantum systems: where the bath Hamiltonian goes

This section answers the question your reader raised: *if the qubit is coupled to a bath, where is the qubit–bath interaction Hamiltonian in the equations, and why does it disappear?* The short answer: it is still there microscopically, but after we (i) enlarge the description to system+bath, (ii) assume weak coupling and short bath memory, and (iii) trace the bath out, the interaction Hamiltonian is *converted* into the dissipator $\mathcal{L}[\rho]$ — a term acting on the system alone. We now show this explicitly.

2.1 Why a closed qubit never thermalizes

A closed quantum system evolves unitarily, $\rho(t) = U(t)\rho(0)U^\dagger(t)$ with $U = e^{-iHt}$. For the single qubit (4), $[\rho, H]$ generates only phases: the populations $\langle 0|\rho|0\rangle$, $\langle 1|\rho|1\rangle$ are constant in time. A closed qubit therefore *cannot* relax to the Gibbs state (5); it just keeps whatever populations it started with. Thermalization, dissipation, and irreversibility are impossible without an environment. This is the physical reason a bath is not optional: it is the only object in the model that can change populations and drive the system to a steady state.

2.2 The microscopic system+bath+interaction Hamiltonian

We restore the environment explicitly. Model the bath as a large collection of harmonic oscillators (the standard Caldeira–Leggett / quantum–optics reservoir),

$$H_B = \sum_k \omega_k a_k^\dagger a_k, \quad (9)$$

with bosonic modes $[a_k, a_{k'}^\dagger] = \delta_{kk'}$. The qubit couples to the bath *bilinearly and transversally*,

$$H_{SB} = \sigma_x \otimes B, \quad B := \sum_k \lambda_k (a_k + a_k^\dagger), \quad (10)$$

where B is a bath “force” operator and λ_k are coupling constants. The *total* Hamiltonian on $\mathcal{H}_S \otimes \mathcal{H}_B$ is

$$H_{\text{tot}} = \underbrace{H_S}_{\epsilon n} + \underbrace{H_B}_{\sum_k \omega_k a_k^\dagger a_k} + \underbrace{H_{SB}}_{\sigma_x \otimes B}. \quad (11)$$

Two modelling choices deserve a word, because they are exactly the places where physics enters.

Approximation — bilinear coupling, (10)

Where: the system–bath term is taken linear in the bath operators, $H_{SB} = \sigma_x \otimes \sum_k \lambda_k (a_k + a_k^\dagger)$. **Why:** this is the leading term in an expansion of a generic coupling in bath displacements; it is *exact* for a Gaussian (harmonic) environment. **When:** valid whenever the environment responds linearly (weak, non–saturating coupling), which is the regime of local detailed balance we ultimately need.

Remark 2.1 (why σ_x and not σ_z). The qubit operator in (10) is *transverse* ($\sigma_x = \sigma_+ + \sigma_-$). It must be, because to thermalize the bath has to be able to *flip* the qubit between $|0\rangle$ and $|1\rangle$, exchanging the energy quantum ϵ . A longitudinal coupling $\sigma_z \otimes B$ commutes with H_S and only randomizes the phase (pure dephasing): it can never change populations and thus never produces a thermal state. Keep this contrast in mind — when we later add a Floquet drive we will deliberately make it *longitudinal*, precisely so that it does *not* pump populations.

2.3 Tracing out the bath: the Born–Markov–secular derivation

We want an equation for the reduced system state $\rho(t) := \text{Tr}_B \rho_{\text{tot}}(t)$ alone. Move to the interaction picture with respect to $H_0 := H_S + H_B$ (tilde denotes interaction picture),

$$\tilde{H}_{SB}(t) = e^{iH_0 t} H_{SB} e^{-iH_0 t} = \tilde{\sigma}_x(t) \otimes \tilde{B}(t). \quad (12)$$

The exact (von Neumann) equation $\dot{\tilde{\rho}}_{\text{tot}} = -i[\tilde{H}_{SB}(t), \tilde{\rho}_{\text{tot}}]$ is integrated once and resubstituted, giving the exact integro–differential equation

$$\dot{\tilde{\rho}}_{\text{tot}}(t) = -i[\tilde{H}_{SB}(t), \rho_{\text{tot}}(0)] - \int_0^t ds [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(s), \tilde{\rho}_{\text{tot}}(s)]]. \quad (13)$$

Tracing over the bath and applying three successive, clearly labelled approximations turns this into a time-local master equation.

Approximation — (A1) Born / weak coupling

Where: we assume the state stays factorized, $\rho_{\text{tot}}(s) \approx \rho(s) \otimes \rho_B$, with ρ_B the fixed thermal state of the bath. **Why:** the coupling λ_k is weak, so system-bath correlations built up are $O(\lambda^2)$ and negligible in the leading dissipator. **When:** λ (equivalently the induced rate γ) much smaller than the system gap ϵ and the bath bandwidth. Choosing ρ_B thermal and $\text{Tr}_B[B\rho_B] = 0$ kills the first term of (13).

Approximation — (A2) Markov / short bath memory

Where: replace $\tilde{\rho}(s) \rightarrow \tilde{\rho}(t)$ in the integral and send the lower limit to $-\infty$. **Why:** the bath correlation function $C(\tau) = \text{Tr}_B[\tilde{B}(\tau)B\rho_B]$ decays on a microscopic time τ_c much shorter than the system's relaxation time, so the system has no memory of its past on the scale that matters. **When:** $\tau_c \ll 1/\gamma$. This is the step that makes the equation *time-local* (memoryless).

After (A1)–(A2), tracing out the bath yields the Redfield equation. Its rates are the one-sided Fourier transforms of the bath correlation function evaluated at the qubit's Bohr frequency $\pm\epsilon$:

$$\gamma(\omega) := \int_{-\infty}^{\infty} d\tau e^{i\omega\tau} C(\tau) = 2\pi J(\omega) [1 + n_B(\omega)], \quad (14)$$

where $J(\omega) = \sum_k \lambda_k^2 \delta(\omega - \omega_k)$ is the *spectral density* and n_B the Bose occupation. The last, purely mathematical, step secures complete positivity.

Approximation — (A3) secular / rotating-wave

Where: drop terms rotating at $e^{\pm 2i\epsilon t}$ in the interaction-picture dissipator. **Why:** they oscillate fast compared to the slow relaxation γ and average to zero over the relaxation time. **When:** $\gamma \ll \epsilon$ (well-resolved gap). This step guarantees the generator is of Lindblad (GKSL) form and hence completely positive.

The result, back in the Schrödinger picture, is the GKSL master equation

$$\dot{\rho} = -i[H_S, \rho] + \gamma_{\downarrow} \left(\sigma_- \rho \sigma_+ - \frac{1}{2} \{ \sigma_+ \sigma_-, \rho \} \right) + \gamma_{\uparrow} \left(\sigma_+ \rho \sigma_- - \frac{1}{2} \{ \sigma_- \sigma_+, \rho \} \right). \quad (15)$$

This is the promised structure $\dot{\rho} = -i[H, \rho] + \mathcal{L}[\rho]$ of [paper Eq. (1)]. **Here is the answer to the reader's question:** the interaction Hamiltonian $H_{SB} = \sigma_x \otimes B$ is *no longer present as a Hamiltonian*. It has been converted, by the elimination of the bath, into the two dissipative channels of (15):

- $\gamma_{\downarrow}$ multiplies σ_- (emission: the qubit gives a quantum ϵ to the bath);
- $\gamma_{\uparrow}$ multiplies σ_+ (absorption: the qubit takes a quantum ϵ from the bath).

The bath Hamiltonian H_B has disappeared entirely; the coupling H_{SB} survives only through the two numbers $\gamma_{\downarrow}, \gamma_{\uparrow}$. *That is why* the working equations of the paper contain a free Hamiltonian H_S and a dissipator $\mathcal{L}$, but no qubit-bath interaction Hamiltonian: the latter was spent to build $\mathcal{L}$.

2.4 Detailed balance fixes the ratio of the rates

The two rates are not independent. The Kubo–Martin–Schwinger (KMS) condition of a thermal bath forces $\gamma(-\omega) = e^{-\beta\omega}\gamma(\omega)$; applied to $\omega = \epsilon$,

$$\frac{\gamma_{\uparrow}}{\gamma_{\downarrow}} = e^{-\beta\epsilon}. \quad (16)$$

This is *local detailed balance*. Its whole purpose is to guarantee that the steady state of (15) reproduces the Boltzmann ratio (8). Indeed, writing $p := \langle 1 | \rho | 1 \rangle$, the populations obey the classical rate equation

$$\dot{p} = \gamma_{\uparrow}(1 - p) - \gamma_{\downarrow}p, \quad (17)$$

whose fixed point $\dot{p} = 0$ is

$$p_{\text{ss}} = \frac{\gamma_{\uparrow}}{\gamma_{\uparrow} + \gamma_{\downarrow}} = \frac{e^{-\beta\epsilon}}{1 + e^{-\beta\epsilon}} = g(\beta\epsilon). \quad (18)$$

The qubit relaxes to exactly the thermal excited population (7), as it must.

Summary of Sec. 2. Coupling a qubit to a bath means writing $H_{\text{tot}} = H_S + H_B + H_{SB}$; eliminating the bath under weak coupling (A1), short memory (A2) and secular averaging (A3) turns H_{SB} into a dissipator $\mathcal{L}[\rho]$ acting on the system alone. The bath’s only fingerprints in the final equation are the two rates, whose ratio is pinned by detailed balance $\gamma_{\uparrow}/\gamma_{\downarrow} = e^{-\beta\epsilon}$ so that the steady state is the Gibbs state. This is why the paper never writes an explicit qubit–bath interaction Hamiltonian.

2.5 The reset model: a minimal thermalization dissipator

Equation (15) is more detail than the logic of the gate needs. Since only the *populations* matter (the steady states are diagonal), the paper adopts the simplest dissipator with the correct fixed point: the *reset model* [paper, Sec. II A 3],

$$\mathcal{L}^{(k)}[\rho] = \gamma_k \left(\text{Tr}_k[\rho] \otimes \tau(\beta_k) - \rho \right). \quad (19)$$

Its interpretation is collisional: in each infinitesimal interval, qubit k has probability $\gamma_k dt$ of being “reset” — swapped for a fresh qubit drawn from the bath’s thermal state $\tau(\beta_k)$ — and probability $1 - \gamma_k dt$ of being left alone. $\text{Tr}_k[\rho]$ is the partial trace over qubit k (the rest of the machine), so $\text{Tr}_k[\rho] \otimes \tau(\beta_k)$ is the state *after* a full thermalization event on qubit k .

Approximation — (A4) reset thermalization, (19)

Where: we replace the microscopically derived dissipator (15) by the reset form (19). **Why:** both share the same fixed point (the local Gibbs state, $\mathcal{L}^{(k)}[\tau(\beta_k)] = 0$) and the same detailed balance; the reset model just fixes the relaxation to be a single rate γ_k for all observables. **When:** whenever only steady–state populations and their thermodynamics are needed — which is the whole computation. Changing the thermalization model changes the exact nonlinear shape of the transfer curve but not the machine’s ability to invert temperatures (this is stated explicitly around [paper Eq. (23)]).

For a single qubit, (19) reduces to $\mathcal{L}[\rho] = \gamma(\tau(\beta) - \rho)$, whose population equation is $\dot{p} = \gamma[g(\beta\epsilon) - p]$ — the same fixed point as before, with relaxation rate γ .

2.6 Heat current

The heat delivered from a qubit to its bath is the energy the dissipator removes from the system,

$$j_k := \text{Tr}[H \mathcal{L}^{(k)}[\rho]]. \quad (20)$$

This is [paper Eq. (5)]. For the reset model (19) with a single qubit of gap ϵ_k and excited population p_k ,

$$\begin{aligned} j_k &= \text{Tr}[\epsilon_k n \gamma_k (\tau(\beta_k) - \rho)] = \gamma_k \epsilon_k (\langle 1 | \tau(\beta_k) | 1 \rangle - p_k) \\ &= \boxed{\gamma_k \epsilon_k [g(\beta_k \epsilon_k) - p_k]}. \end{aligned} \quad (21)$$

This is [paper Eq. (6)]. The sign convention: $j_k > 0$ when the bath pumps energy into the qubit ($g(\beta_k \epsilon_k) > p_k$, the qubit is colder than its bath). A qubit coupled to *two* baths simply gets the sum of two such dissipators, and the total current is the sum — a fact we will use for the modulator.

How to implement — the reset dissipator and current

In QuTiP the reset dissipator is most easily built as a rate times the difference of superoperators, or, for populations only, integrated as the classical rate equation (17). For a full Lindblad simulation one instead uses collapse operators $\{\sqrt{\gamma_\downarrow} \sigma_-, \sqrt{\gamma_\uparrow} \sigma_+\}$ with $\gamma_\downarrow = \gamma(1 - g)$, $\gamma_\uparrow = \gamma g$, $g = g(\beta\epsilon)$; `mesolve` then reproduces (21) as `expect(H, rho)`'s rate of change due to that bath.

3 Multi-qubit thermal machines

3.1 Free and interaction Hamiltonian

A machine is several qubits, labelled k , each with gap ϵ_k . The free Hamiltonian is the sum

$$H_0 = \sum_k \epsilon_k |1\rangle \langle 1|_k, \quad (22)$$

where $|1\rangle \langle 1|_k$ acts as identity on all other qubits. The qubits are coupled by a *time-independent, energy-preserving* interaction H_{int} , meaning

$$[H_{\text{int}}, H_0] = 0. \quad (23)$$

Equation (23) is a strong and deliberate constraint. It says the interaction can only connect *degenerate* eigenstates of H_0 — it shuffles excitations among qubits without changing the total bare energy. We will see in Sec. 5 that this single requirement, together with a resonance condition on the gaps, *uniquely* dictates the form of the collector's three-body term.

Approximation — (A5) energy-preserving interaction, (23)

Where: $[H_{\text{int}}, H_0] = 0$. **Why:** in the weak internal-coupling regime, any non-commuting (energy-changing) part of the interaction is off-resonant and averages away under the same rotating-wave logic as (A3); only the energy-conserving, secular part survives to influence the steady state. **When:** coupling $\chi \ll$ the gaps, so that off-resonant transitions are suppressed by χ/ϵ . This is what justifies the local master equation (24) below.

3.2 The local master equation

Each qubit k is weakly coupled to its own bath B_k at inverse temperature β_k . Under weak coupling the machine obeys the *local* master equation

$$\dot{\rho} = -i[H_0 + H_{\text{int}}, \rho] + \sum_k \mathcal{L}^{(k)}[\rho], \quad (24)$$

with each $\mathcal{L}^{(k)}$ the reset dissipator (19) for its own bath. This is [paper Eq. (3)]. “Local” means each dissipator is built from the *bare* qubit operators, not the dressed eigenstates of $H_0 + H_{\text{int}}$.

Approximation — (A6) local vs. global master equation

Where: the dissipators in (24) act on individual bare qubits. **Why:** when H_{int} is weak compared with the gaps, the eigenstates of $H_0 + H_{\text{int}}$ are close to the bare product states, and the local dissipator is both simpler and thermodynamically consistent. **When:** $\chi \ll \epsilon_k$; in the opposite (strong internal coupling) regime one must use the global dissipator built from the dressed eigenoperators, otherwise small violations of the second law can appear (Hofer *et al.*, cited as [51] in the paper). We stay local throughout.

3.3 Local detailed balance and heat currents

The key property inherited from Sec. 2 is that each local dissipator has the local Gibbs state as a zero,

$$\mathcal{L}^{(k)}[\tau(\beta_k)] = 0, \quad (25)$$

which is [paper Eq. (4)]. The heat current into bath k is again (20), and, crucially, in a nonequilibrium steady state (NESS) the machine’s own entropy is constant, so all the entropy the machine produces is dumped into the baths.

3.4 Entropy production and the second law

Define the machine’s von Neumann entropy $S(\rho) = -\text{Tr}[\rho \ln \rho]$. The entropy production rate is

$$\dot{\Sigma} := \dot{S}(\rho(t)) - \sum_k \beta_k j_k(t) \geq 0, \quad (26)$$

which is [paper Eq. (7)] and is the second law for this class of machines. In the steady state $\dot{S} = 0$, so $\dot{\Sigma} = -\sum_k \beta_k j_k \geq 0$ is purely the weighted heat dumped into the environments. This quantity is the “dissipation” whose trade-off against computational accuracy we quantify in Part III. It is also the microscopic meaning of the “information lost to the environment” discussed qualitatively in the Floquet-buffer note.

End of Part I. We now have every generic tool: a free Hamiltonian $H_0 = \sum_k \epsilon_k n_k$, an energy-preserving interaction H_{int} (still to be constructed), a local reset dissipator $\sum_k \mathcal{L}^{(k)}$ with detailed balance, a heat current $j_k = \gamma_k \epsilon_k [g(\beta_k \epsilon_k) - p_k]$, and an entropy production $\dot{\Sigma} \geq 0$. Part II uses two qubits to build the virtual qubit; Part III assembles the NOT gate.

Part II

The virtual qubit and the resonant interaction

4 Two thermal qubits and the virtual qubit

4.1 The product thermal state

Take two qubits C_0, C_1 with gaps ϵ_0 and $\epsilon_1 \leq \epsilon_0$, each in contact with its own bath at inverse temperature β_0, β_1 . In the absence of any interaction, each qubit independently reaches its Gibbs state (6),

$$\tau_{C_i}(\beta_i) = \frac{1}{Z_{C_i}} \left(|0\rangle \langle 0|_{C_i} + e^{-\beta_i \epsilon_i} |1\rangle \langle 1|_{C_i} \right), \quad Z_{C_i} = 1 + e^{-\beta_i \epsilon_i}, \quad (27)$$

[paper Eq. (8)]. The joint state is the product $\tau_{C_0}(\beta_0) \otimes \tau_{C_1}(\beta_1)$, diagonal in the four states $|i\rangle_{C_0} |j\rangle_{C_1}$, $i, j \in \{0, 1\}$, with populations

$$\begin{aligned} P_{00} &= p_0(0) p_1(0), & P_{01} &= p_0(0) p_1(1), \\ P_{10} &= p_0(1) p_1(0), & P_{11} &= p_0(1) p_1(1), \end{aligned} \quad (28)$$

where $P_{ij} := \langle ij | \tau_{C_0} \otimes \tau_{C_1} | ij \rangle$ is the joint population of $|i\rangle_{C_0} |j\rangle_{C_1}$, and $p_i(1) = g(\beta_i \epsilon_i)$, $p_i(0) = 1 - p_i(1)$.

4.2 The virtual qubit subspace

Focus on the two “crossed” states

$$|0\rangle_v := |0\rangle_{C_0} |1\rangle_{C_1}, \quad |1\rangle_v := |1\rangle_{C_0} |0\rangle_{C_1}. \quad (29)$$

This is [paper Eq. (15)] (collector convention). They span a two-dimensional subspace of the four-dimensional joint space — a *virtual qubit*. Its energy gap is the difference of the bare energies of the two states,

$$\epsilon_v := E(|1\rangle_v) - E(|0\rangle_v) = \epsilon_0 - \epsilon_1. \quad (30)$$

The name “virtual” is literal: no physical two-level system with gap ϵ_v exists; it is an effective qubit encoded in a corner of the joint Hilbert space of C_0 and C_1 .

4.3 The virtual temperature

Assign the virtual qubit a temperature exactly as we did for a real one, via its population ratio (8). Using (28) and the Boltzmann ratios $p_i(1)/p_i(0) = e^{-\beta_i \epsilon_i}$,

$$e^{-\beta_v \epsilon_v} := \frac{\langle 1 |_v \tau_{C_0} \otimes \tau_{C_1} | 1 \rangle_v}{\langle 0 |_v \tau_{C_0} \otimes \tau_{C_1} | 0 \rangle_v} = \frac{P_{10}}{P_{01}} = \frac{p_0(1)p_1(0)}{p_0(0)p_1(1)} = \frac{e^{-\beta_0 \epsilon_0}}{e^{-\beta_1 \epsilon_1}}. \quad (31)$$

This is [paper Eq. (10)]. Taking logarithms, $\beta_v \epsilon_v = \beta_0 \epsilon_0 - \beta_1 \epsilon_1$, and dividing by $\epsilon_v = \epsilon_0 - \epsilon_1$,

$$\boxed{\beta_v = \frac{\epsilon_0}{\epsilon_0 - \epsilon_1} \beta_0 - \frac{\epsilon_1}{\epsilon_0 - \epsilon_1} \beta_1.} \quad (32)$$

This is [paper Eq. (11)/(16)]. The virtual (inverse) temperature is a *linear combination of the input inverse temperatures*, with weights set by the energy gaps. *This is the weighted sum of the perceptron*, and it is the reason the machine computes. Note that β_v can exceed both β_0, β_1 , lie between them, or go negative — the three regimes below.

4.4 Refrigerator, heat pump, heat engine

Holding β_0 fixed and varying β_1 in (32):

- $\beta_1 < \beta_0$ (hot input): β_v is larger than both β_0 and β_1 — the virtual qubit is *colder* than either bath; coupling a target to it *cools* the target (refrigerator).
- $\beta_0 < \beta_1 < (\epsilon_0/\epsilon_1)\beta_0$: β_v is smaller than both — *hotter* virtual qubit (heat pump).
- $\beta_1 > (\epsilon_0/\epsilon_1)\beta_0$: $\beta_v < 0$ — population inversion (heat engine).

These are the regions of [paper Fig. 2]. The ability of a single knob β_1 to move the target across cooling/heating is the physical basis of temperature inversion, i.e. of the NOT gate.

How to implement — virtual temperature

Directly: $\text{beta_v} = (\text{eps0}/(\text{eps0}-\text{eps1})) * \text{beta0} - (\text{eps1}/(\text{eps0}-\text{eps1})) * \text{beta1}$. As a check, build $\text{tau01} = \text{tensor}(\text{thermal}(\text{beta0}, \text{eps0}), \text{thermal}(\text{beta1}, \text{eps1}))$ and verify (31): the ratio of the $|10\rangle$ and $|01\rangle$ diagonal entries equals $e^{-\beta_v(\epsilon_0 - \epsilon_1)}$.

5 Adding the target qubit: the resonant three-body interaction

We now derive — not quote — the collector’s interaction Hamiltonian, the object your reader most wants to see written out and justified.

5.1 The resonance condition

Introduce a third qubit C_z with gap ϵ_z , intended to be placed in “thermal contact” with the virtual qubit so that it inherits β_v . For a weak, energy-preserving interaction to move excitations between the virtual qubit and C_z , the two must be *resonant*: the energy released by de-exciting the virtual qubit must exactly excite C_z . Hence we *choose*

$$\boxed{\epsilon_z = \epsilon_v = \epsilon_0 - \epsilon_1.} \quad (33)$$

This is a design equation, imposed by hand on the fabricated gaps [paper, above Eq. (14)]. Its consequence: the two joint states

$$|1\rangle_v |0\rangle_{C_z} = |1\rangle_{C_0} |0\rangle_{C_1} |0\rangle_{C_z} = |100\rangle, \quad |0\rangle_v |1\rangle_{C_z} = |0\rangle_{C_0} |1\rangle_{C_1} |1\rangle_{C_z} = |011\rangle, \quad (34)$$

become *degenerate*. Their bare energies are

$$E(|100\rangle) = \epsilon_0, \quad E(|011\rangle) = \epsilon_1 + \epsilon_z \stackrel{(33)}{=} \epsilon_0. \quad (35)$$

(Ordering of kets is $(C_0 C_1 C_z)$ throughout.)

5.2 Deriving the interaction from $[H_{\text{int}}, H_0] = 0$

The energy-preserving requirement (23) says H_{int} may only connect degenerate eigenstates of H_0 . Among all three-qubit basis states, the single resonant pair created by (33) is $\{|100\rangle, |011\rangle\}$. The most general Hermitian operator connecting *only* this pair is

$$\boxed{H_{\text{int}} = \chi(|100\rangle\langle 011| + |011\rangle\langle 100|),} \quad (36)$$

with a single real coupling χ (a phase on χ can be absorbed into the definition of the states). Written with Pauli operators, using $|100\rangle\langle 011| = \sigma_+^{(0)} \sigma_-^{(1)} \sigma_-^{(z)}$,

$$H_{\text{int}} = \chi \left(\sigma_+^{(0)} \sigma_-^{(1)} \sigma_-^{(z)} + \sigma_-^{(0)} \sigma_+^{(1)} \sigma_+^{(z)} \right), \quad (37)$$

or, in the mixed notation of [paper Eq. (14)],

$$H_{\text{int}} = \chi |1\rangle \langle 0|_{C_0} \otimes |0\rangle \langle 1|_{C_1} \otimes |0\rangle \langle 1|_{C_z} + \text{h.c.} \quad (38)$$

Equations (36)–(38) are one and the same operator; (38) is verbatim the paper’s. One checks directly that $[H_{\text{int}}, H_0] = 0$ because it connects only degenerate states.

5.3 Why this term, and why not the others

Five questions, five answers.

- (i) **Why does energy conservation force exactly this pair?** Because (33) makes $|100\rangle$ and $|011\rangle$ the *only* degenerate pair differing by a single allowed excitation rearrangement; any other off-diagonal term would connect states of different bare energy and violate (23).
- (ii) **Why three-body and not pairwise?** The virtual qubit (29) is a *joint* property of C_0 and C_1 . A two-body term $\sigma_+^{(0)}\sigma_-^{(1)}$ merely rotates the excitation between C_0 and C_1 (an internal rotation of the virtual qubit) and transfers nothing to C_z . To let C_z “read” the virtual qubit you must flip all three spins at once — an irreducibly three-body operator.
- (iii) **What does it do physically?** Rewrite (36) in virtual-qubit variables:

$$H_{\text{int}} = \chi (|1\rangle_v |0\rangle_{C_z} \langle 0|_v \langle 1|_{C_z} + \text{h.c.}), \quad (39)$$

[paper Eq. (12)] — a resonant SWAP (beam splitter) between the virtual qubit and C_z . *This is the microscopic reason C_z thermalizes to β_v :* the interaction literally swaps the virtual qubit’s population into C_z , so in steady state C_z carries the virtual population, i.e. the virtual temperature.

- (iv) **Why drop the counter-rotating / non-resonant terms?** Operators like $\sigma_+^{(0)}\sigma_+^{(1)}\sigma_+^{(z)}$ create three excitations and oscillate at $\epsilon_0 + \epsilon_1 + \epsilon_z$ in the interaction picture; being far off resonance they average to zero on the relaxation time (the same RWA as (A3)/(A5)), with error $O(\chi/\epsilon)$.
- (v) **Why is there no bare C_0 – C_1 coupling or $\sigma_z\sigma_z$ term?** Such terms would hybridize the two input qubits and spoil the *product* thermal structure $\tau_{C_0} \otimes \tau_{C_1}$ from which β_v was defined in (31). They are deliberately absent. The three lines a schematic draws between C_0, C_1, C_z represent the single three-body term (36), not three independent couplings.

Remark 5.1 (a frequent coding error). The resonant pair is $|100\rangle \leftrightarrow |011\rangle$. A tempting but *wrong* choice is $|101\rangle \leftrightarrow |010\rangle$, whose bare energies are $\epsilon_0 + \epsilon_z$ and ϵ_1 ; these differ by $\epsilon_0 + \epsilon_z - \epsilon_1 = 2\epsilon_z \neq 0$. That term is off-resonant and, at $\chi \ll \epsilon_z$, dynamically inert — the collector would not work. Always build H_{int} from the degenerate pair fixed by (33).

How to implement — the collector Hamiltonian

With QuTiP and ordering $(C_0 C_1 C_z)$:

```
n0,n1,nz = num on each site (tensor with identities)
H0 = eps0*n0 + eps1*n1 + eps_z*nz
k100 = tensor(basis(2,1),basis(2,0),basis(2,0))
k011 = tensor(basis(2,0),basis(2,1),basis(2,1))
Hint = chi*(k100*k011.dag() + k011*k100.dag())
H = H0 + Hint
```

Verify $\|[H_0, H_{\text{int}}]\| = 0$ before trusting anything: it is the one-line check that the resonance (33) was implemented correctly.

5.4 Steady state of the swap: $C_z \rightarrow \beta_v$

With C_z additionally reset-coupled (rate μ) to a bath B_z at β_z , the competition is between the swap (39), which pulls C_z toward β_v , and the bath, which pulls it toward β_z . Under the time-scale separation introduced next (Part III), the current the collector delivers to B_z is [paper Eq. (17)]

$$j_C = \mu \epsilon_z [g_z(\beta_z) - g_z(\beta_v)], \quad g_z(x) := g(x \epsilon_z). \quad (40)$$

When $\beta_v > \beta_z$ the collector cools B_z (refrigerator); when $\beta_v < \beta_z$ it heats it (heat pump). Setting $j_C = 0$ in isolation gives $\beta_z = \beta_v$: *the collector alone drives the output reservoir to the virtual temperature*. That linearity (cf. (32)) is precisely why the collector by itself is only an inverting *linear amplifier* — the motivation for the modulator, taken up in Part III.

End of Part II. The collector's Hamiltonian is $H_C = H_0 + H_{\text{int}}$ with $H_0 = \epsilon_0 n_0 + \epsilon_1 n_1 + \epsilon_z n_z$ and the resonant three-body swap (36). It thermalizes C_z (and hence B_z) to the virtual temperature β_v of (32), a weighted sum of the inputs — the perceptron pre-activation. Part III adds the modulator to make the response nonlinear and reads out the bit.

Part III

The NOT gate: collector, modulator, dynamics, logic

6 The collector alone is a linear amplifier

Assemble the collector: three qubits C_0, C_1, C_z with the free Hamiltonian $H_0 = \epsilon_0 n_0 + \epsilon_1 n_1 + \epsilon_z n_z$, the resonant three-body interaction (36), and three local reset dissipators (rates γ for C_0, C_1 ; μ for C_z). Its master equation is [paper Eq. (13)]

$$\dot{\rho}_C = -i[H_0 + H_{\text{int}}, \rho_C] + \mathcal{L}^{(0)}[\rho_C] + \mathcal{L}^{(1)}[\rho_C] + \mathcal{L}^{(z)}[\rho_C]. \quad (41)$$

As shown in Sec. 5, the swap thermalizes C_z toward the virtual temperature β_v , delivering to B_z the current (40), $j_C = \mu \epsilon_z [g_z(\beta_z) - g_z(\beta_v)]$. In isolation the fixed point $j_C = 0$ is $\beta_z = \beta_v$, and by (32) this is an *affine* (linear) function of the input:

$$\beta_z^{\text{collector only}} = \beta_v = \frac{\epsilon_0}{\epsilon_z} \beta_0 - \frac{\epsilon_1}{\epsilon_z} \beta_1. \quad (42)$$

The output inverse temperature is a straight line in β_1 with negative slope: an *inverting linear amplifier*.

7 Why we need the modulator

Here is the answer to the question “why do we even need the modulator?”. A linear amplifier is a *bad* logic gate for one concrete reason: it does not restore levels. Suppose the input carries a small fluctuation $\beta_1 \rightarrow \beta_1 + \delta$. By (42) the output moves by $-(\epsilon_1/\epsilon_z)\delta$; with ϵ_1/ϵ_z made large (needed for a steep, decisive gate) the *noise is amplified* along with the signal. Cascading such gates amplifies noise without bound, causing bit flips. A good gate must instead *saturate*: for inputs safely inside a logical region the output should barely move (flat response, noise rejected), while it should swing sharply when the input crosses the decision boundary. That is precisely a *sigmoidal*, not linear, transfer curve.

The role of the modulator (one sentence). The collector supplies the perceptron’s *weighted sum* β_v (32); the modulator supplies the perceptron’s *nonlinear activation* by confining the output to a fixed logical window $[\beta_{\text{hot}}, \beta_{\text{cold}}]$ and thereby saturating the response. Without it the “activation function” is the identity $f(y) = y$, which is known to make a useless classifier.

8 The modulator: one qubit, two baths

The modulator $\mathcal{M}$ is a *single* qubit $C_{\mathcal{M}}$ with gap chosen resonant with the output channel,

$$H_{\mathcal{M}} = \epsilon_{\mathcal{M}} n_{\mathcal{M}}, \quad \epsilon_{\mathcal{M}} = \epsilon_z, \quad (43)$$

placed in simultaneous contact with *two* baths: a reference bath B_r at fixed β_r (rate γ) and the output bath B_z (rate μ'). Its state obeys [paper Eq. (18)]

$$\dot{\rho}_{\mathcal{M}} = \mathcal{L}^{(r)}[\rho_{\mathcal{M}}] + \mathcal{L}^{(z)}[\rho_{\mathcal{M}}] = \gamma(\tau(\beta_r) - \rho_{\mathcal{M}}) + \mu'(\tau(\beta_z) - \rho_{\mathcal{M}}). \quad (44)$$

Both dissipators are diagonal, so it suffices to track the excited population $p_{\mathcal{M}} := \langle 1 | \rho_{\mathcal{M}} | 1 \rangle$:

$$\dot{p}_{\mathcal{M}} = \gamma[g_z(\beta_r) - p_{\mathcal{M}}] + \mu'[g_z(\beta_z) - p_{\mathcal{M}}]. \quad (45)$$

Setting $\dot{p}_{\mathcal{M}} = 0$,

$$p_{\mathcal{M}}^{\text{ss}} = \frac{\gamma g_z(\beta_r) + \mu' g_z(\beta_z)}{\gamma + \mu'} \xrightarrow{\mu' \ll \gamma} g_z(\beta_r). \quad (46)$$

Approximation — (A7) modulator pinned to the reference, (46)

Where: we set $\mu' \ll \gamma$, so $p_{\mathcal{M}}^{\text{ss}} \approx g_z(\beta_r)$. **Why:** the strong reference coupling holds the modulator qubit at β_r , essentially unperturbed by the (weak) output coupling. **When:** $\mu'/\gamma \rightarrow 0$; corrections are $O(\mu'/\gamma)$. This is what makes the modulator a stiff “anchor” whose only slow action is to bleed a controlled current into B_z .

The steady-state heat current the modulator delivers to B_z is (21) with $p_{\mathcal{M}} = g_z(\beta_r)$, [paper Eq. (19)]

$$j_{\mathcal{M}} = \mu' \epsilon_z [g_z(\beta_z) - g_z(\beta_r)]. \quad (47)$$

The modulator therefore continuously pulls β_z toward β_r with strength μ' : a single-qubit *heat valve* that competes with the collector current (40).

9 Dynamics of the full machine: two time scales

The output reservoir B_z has finite heat capacity $\mathcal{C}$, so its temperature $T_z = 1/\beta_z$ moves under the total incoming current $j_{\mathcal{C}} + j_{\mathcal{M}}$ (calorimetric law $\mathcal{C}\dot{T}_z = j_{\mathcal{C}} + j_{\mathcal{M}}$). In terms of $\beta_z = 1/T_z$ (units $k_B = 1$),

$$\dot{\beta}_z = -\frac{\beta_z^2}{\mathcal{C}}(j_{\mathcal{C}} + j_{\mathcal{M}}). \quad (48)$$

This is [paper Eq. (20)]. The consistency of inserting the *steady-state* currents (40), (47) into (48) rests on a separation of time scales built into the design,

$$\gamma \gg \mu, \mu'. \quad (49)$$

Approximation — (A8) adiabatic elimination / time-scale separation, (49)

Where: we use the *steady-state* $C_z \rightarrow \beta_v$ and $C_{\mathcal{M}} \rightarrow \beta_r$ inside the slow equation (48).
Why: with $\gamma \gg \mu, \mu'$ the internal qubits relax fast compared with the drift of $\beta_z(t)$, so at each instant they are slaved to the current value of β_z . **When:** $\mu, \mu' \ll \gamma$; the error is $O(\mu/\gamma)$. This is the autonomous analogue of the quasi-static assumption in circuit analysis, and it is what lets us treat $\beta_z(t)$ as the single slow logical variable.

10 Steady state and the transfer characteristic

The output settles when $\dot{\beta}_z = 0$, i.e. $j_C + j_{\mathcal{M}} = 0$. Substituting (40) and (47),

$$\mu[g_z(\beta_z) - g_z(\beta_v)] + \mu'[g_z(\beta_z) - g_z(\beta_r)] = 0, \quad (50)$$

so, with the branching ratio $\Delta := \mu/(\mu + \mu')$,

$$g_z(\beta_z^\infty) = \Delta g_z(\beta_v) + (1 - \Delta) g_z(\beta_r). \quad (51)$$

This is [paper Eq. (21)]. The output excitation is a convex combination of the collector's target (β_v) and the modulator's anchor (β_r), weighted purely by coupling rates.

10.1 Fixing the modulator: confinement to the logical window

For β_z^∞ to be a valid logical signal we demand it stay in a window $[\beta_{\text{hot}}, \beta_{\text{cold}}]$ ($\beta_{\text{cold}} > \beta_{\text{hot}}$) even when the virtual temperature saturates. Since g_z is monotonically decreasing, $g_z(\beta_v) \rightarrow 0$ as $\beta_v \rightarrow +\infty$ and $g_z(\beta_v) \rightarrow 1$ as $\beta_v \rightarrow -\infty$. Impose the two rail conditions on (51):

$$\beta_v \rightarrow +\infty : \quad (1 - \Delta)g_z(\beta_r) \stackrel{!}{=} g_z(\beta_{\text{cold}}), \quad (52)$$

$$\beta_v \rightarrow -\infty : \quad \Delta + (1 - \Delta)g_z(\beta_r) \stackrel{!}{=} g_z(\beta_{\text{hot}}). \quad (53)$$

Subtracting (52) from (53) gives the two design equations that fix the modulator's free parameters:

$$\Delta = g_z(\beta_{\text{hot}}) - g_z(\beta_{\text{cold}}), \quad g_z(\beta_r) = \frac{g_z(\beta_{\text{cold}})}{1 - \Delta}. \quad (54)$$

These pin $(\mu'/\mu, \beta_r)$ from the chosen logical window. This is the main-text version of [paper Supplementary Material A]; the sign conventions there $(\beta_{\text{min}}, \beta_{\text{max}}, \kappa)$ are identical after relabelling.

10.2 The transfer function

Substituting (54) back into (51), $(1 - \Delta)g_z(\beta_r) = g_z(\beta_{\text{cold}})$ and $\Delta = g_z(\beta_{\text{hot}}) - g_z(\beta_{\text{cold}})$, so

$$g_z(\beta_z^\infty) = g_z(\beta_{\text{hot}})g_z(\beta_v) + g_z(\beta_{\text{cold}})(1 - g_z(\beta_v)) =: Q(\beta_v). \quad (55)$$

Inverting $g_z(\beta) = (1 + e^{\beta\epsilon_z})^{-1}$ (so $\beta = \epsilon_z^{-1} \ln[g_z^{-1} - 1]$) yields the exact transfer characteristic [paper Eq. (22)]

$$\beta_z^\infty = \frac{1}{\epsilon_z} \ln[Q(\beta_v)^{-1} - 1], \quad Q(\beta_v) = g_z(\beta_{\text{hot}})g_z(\beta_v) + g_z(\beta_{\text{cold}})(1 - g_z(\beta_v)). \quad (56)$$

How to implement — transfer curve

Compute β_{v} from (32), then $g_z(x) = 1/(1 + \exp(x \cdot \epsilon_z))$, $Q = g_z(\beta_{\text{hot}}) \cdot g_z(\beta_{\text{v}}) + g_z(\beta_{\text{cold}}) \cdot (1 - g_z(\beta_{\text{v}}))$, and $\beta_z^{\text{inf}} = \log(1/Q - 1)/\epsilon_z$. Sweeping β_1 reproduces [paper Fig. 3B]. As a cross-check, integrate (48) with a stiff ODE solver and confirm the fixed point matches (56).

11 Emergence of the sigmoid

The link to the perceptron activation is now exact. First note the identity, from $\beta_v \epsilon_z = \beta_0 \epsilon_0 - \beta_1 \epsilon_1$ (multiply (32) by $\epsilon_z = \epsilon_0 - \epsilon_1$),

$$g_z(\beta_v) = \frac{1}{1 + e^{\beta_v \epsilon_z}} = f(\beta_0 \epsilon_0 - \beta_1 \epsilon_1), \quad f(x) := \frac{1}{1 + e^x}, \quad (57)$$

with *no approximation*: the collector’s virtual population is already a sigmoid of the perceptron pre-activation $x = \beta_0 \epsilon_0 - \beta_1 \epsilon_1$. The modulator’s job is only to map it affinely onto the temperature axis. Expanding (56) for small ϵ_z using $g_z(\beta) = \frac{1}{2} - \frac{\beta \epsilon_z}{4} + O(\epsilon_z^3)$:

$$Q(\beta_v) = \frac{1}{2} - \frac{\epsilon_z}{4} B + O(\epsilon_z^2), \quad B := \beta_{\text{hot}} g_z(\beta_v) + \beta_{\text{cold}} (1 - g_z(\beta_v)), \quad (58)$$

so that $\frac{1-Q}{\epsilon_z} = 1 + \epsilon_z B + O(\epsilon_z^2)$ and

$$\beta_z^{\infty} = \frac{1}{\epsilon_z} \ln \frac{1-Q}{Q} = B + O(\epsilon_z) = \beta_{\text{cold}} + (\beta_{\text{hot}} - \beta_{\text{cold}}) g_z(\beta_v) + O(\epsilon_z). \quad (59)$$

Combining (57) and (59),

$$\boxed{\beta_z^{\infty} \simeq \beta_{\text{cold}} + (\beta_{\text{hot}} - \beta_{\text{cold}}) f(\beta_0 \epsilon_0 - \beta_1 \epsilon_1)}, \quad (60)$$

which is [paper Eq. (23)]: a sigmoidal interpolation between the two logical rails whose argument is exactly the perceptron pre-activation. As $\epsilon_1 \rightarrow \infty$ the sigmoid sharpens into the ideal inverted step (perfect NOT), at the cost of dissipation (Sec. 13).

Approximation — (A9) small- ϵ_z (sigmoid) expansion, (59)

Where: we expand g_z and the logarithm to first order in ϵ_z . **Why:** for $\epsilon_z \ll \epsilon_1$ the output gap is the smallest scale and the transfer curve is dominated by its linear term, which is a clean sigmoid. **When:** $\epsilon_z \ll 1/\beta_{\text{hot}, \text{cold}}, \epsilon_1$. For larger ϵ_z the curve still inverts but the parameter roles (β_0 as threshold, ϵ_0 as steepness) blur.

11.1 Threshold and gain

Differentiating (60) with $f'(x) = -f(x)(1 - f(x))$ and $x = \beta_0 \epsilon_0 - \beta_1 \epsilon_1$ (so $dx/d\beta_1 = -\epsilon_1$),

$$\mathcal{A}(\beta_1) := \frac{d\beta_z^{\infty}}{d\beta_1} = -\Delta\beta \epsilon_1 f(x)(1 - f(x)), \quad \Delta\beta := \beta_{\text{cold}} - \beta_{\text{hot}}. \quad (61)$$

The switching threshold is where $x = 0$, i.e. $\beta_1^* = \beta_0 \epsilon_0 / \epsilon_1 = \beta_0(1 + \epsilon_z / \epsilon_1) \approx \beta_0$; there $f(1 - f) = 1/4$ and

$$\mathcal{A}_{\text{max}} = -\frac{\epsilon_1 \Delta\beta}{4}, \quad (62)$$

recovering [paper Supp. A, Eq. (A10)] (for $\beta_{\text{hot}} = 0, \beta_{\text{cold}} = 1, \mathcal{A} = -\epsilon_0/4 + O(\epsilon_z^2)$ with $\epsilon_0 \approx \epsilon_1$). **This makes the trade-off visible already:** steeper gate $\Rightarrow$ larger $\epsilon_1 \Rightarrow$ (Sec. 13) more dissipation. The qubit gap ϵ_1 is the “transconductance” of the device.

12 Logic encoding and readout error

12.1 Encoding and decoding

Logical values are placed on, and read from, temperatures via [paper Eq. (24)]

$$x = \begin{cases} 0, & \beta_1 = \beta_{\text{hot}} \\ 1, & \beta_1 = \beta_{\text{cold}} \end{cases} \quad y = \begin{cases} 0, & \beta_z^\infty \leq (1 + \delta)\beta_{\text{hot}} \\ 1, & \beta_z^\infty \geq (1 - \delta)\beta_{\text{cold}} \\ \emptyset, & \text{otherwise,} \end{cases} \quad (63)$$

where δ is a tolerance (guard band) and $\emptyset$ is an invalid readout. Inversion follows from the sign structure of β_v : a hot input ($\beta_1 = \beta_{\text{hot}} < \beta_0$) makes the collector a refrigerator, $g_z(\beta_v) \rightarrow 0$, so $\beta_z^\infty \rightarrow \beta_{\text{cold}}$ ($y = 1$); a cold input makes it a heat pump, $g_z(\beta_v) \rightarrow 1$, $\beta_z^\infty \rightarrow \beta_{\text{hot}}$ ($y = 0$). That is the NOT gate.

12.2 The Gaussian readout–error model

Because the machine is stochastic, the finite bath fluctuates around β_z^∞ . Model the actual response as Gaussian [paper Eqs. (25)–(26)],

$$T(\beta_z | \beta_1) \propto \mathcal{N}(\beta_z^\infty, C), \quad (64)$$

justified by the central limit theorem for the many–particle reservoir (larger heat capacity $C \Rightarrow$ wider fluctuations). With input distribution $p(x)$, encoding $e(\beta_1 | x)$ and decoding $d(y | \beta_z)$ from (63), the channel is

$$p(y | x) = \int d(y | \beta_z) T(\beta_z | \beta_1) e(\beta_1 | x) d\beta_1 d\beta_z, \quad (65)$$

and the average error is [paper Eq. (27)]

$$\langle \xi \rangle = \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} p(x) p(y | x) \delta_{xy} \quad (\text{probability output} \neq \text{desired}). \quad (66)$$

The sharper the transfer curve (larger ϵ_1), the smaller $\langle \xi \rangle$ — the gate quality improves with energy invested.

13 The error–dissipation trade–off

The dissipation is the entropy production (26) integrated over the run. In the slow regime the machine’s own entropy is constant, so the total rate splits into collector and modulator pieces,

$$\dot{\Sigma} = \dot{\Sigma}_{\mathcal{C}} + \dot{\Sigma}_{\mathcal{M}}, \quad \dot{\Sigma}_{\mathcal{C}} = -\beta_0 j_0 - \beta_1 j_1 - \beta_z j_{\mathcal{C}}, \quad \dot{\Sigma}_{\mathcal{M}} = -\beta_z j_{\mathcal{M}} - \beta_r j_r, \quad (67)$$

with $j_{0,1,r}$ the currents from $B_{0,1,r}$ into their qubits. Integrating over the run time τ and averaging over inputs [paper Eqs. (28)–(29)],

$$\langle \Sigma \rangle = \sum_{x \in \{0,1\}} \int p(x) e(\beta_1 | x) \Sigma(\beta_1) d\beta_1, \quad \Sigma(\beta_1) = \int_0^\tau \dot{\Sigma}(\beta_1) dt. \quad (68)$$

The central trade–off [paper Fig. 3C]. Lowering the average error $\langle \xi \rangle$ requires increasing the gain, i.e. raising ϵ_1 (Eq. (62)); but the heat currents — and hence $\langle \Sigma \rangle$ — grow with the qubit energies. Thus $\langle \xi \rangle$ and $\langle \Sigma \rangle$ are monotonically opposed: *accuracy costs dissipation*.

This is the thermodynamic price of computation for the neuron, and the baseline that the Floquet–buffer extension seeks to improve.

End of Part III. Full machine Hamiltonian: $H = H_0 + H_{\text{int}} + H_{\mathcal{M}}$ (collector free + resonant swap + modulator), plus four local reset baths. The collector builds the weighted sum β_v ; the modulator saturates it into the sigmoid (60); encoding (63) reads the bit; and accuracy trades against dissipation. Part IV lifts this to n inputs and proves the perceptron equivalence.

Part IV

The general neuron: perceptron equivalence and networks

14 The n –input collector

Everything generalizes. The collector now has $n + 2$ qubits: a reference C_0 (bath B_0 , fixed β_0), inputs $C_1, \dots, C_n$ (baths $B_1, \dots, B_n$ at $\beta_1, \dots, \beta_n$ encoding the n input bits), and the output C_z (finite reservoir B_z). The modulator is *exactly the same* single qubit as before. The only new ingredient is that a multi–qubit machine hosts *many* virtual qubits, and we must say which one we use.

14.1 Choosing a virtual qubit: the vector h

Introduce a binary vector $h = (h_0, h_1, \dots, h_n)$, $h_i \in \{0, 1\}$, recording whether qubit i contributes its ground ($h_i = 0$) or excited ($h_i = 1$) state. The chosen virtual qubit has levels [paper Eqs. (30)–(31)]

$$|0\rangle_v := |h_0\rangle_{C_0} |h_1\rangle_{C_1} \cdots |h_n\rangle_{C_n}, \quad (69)$$

$$|1\rangle_v := |h_0 \oplus 1\rangle_{C_0} |h_1 \oplus 1\rangle_{C_1} \cdots |h_n \oplus 1\rangle_{C_n}, \quad (70)$$

where $\oplus$ is addition mod 2: $|1\rangle_v$ is the bit–flip of $|0\rangle_v$ on *every* qubit. Its energy gap is [paper Eq. (32)]

$$\epsilon_v = \sum_{i=0}^n (-1)^{h_i \oplus 1} \epsilon_i. \quad (71)$$

14.2 The interaction and the multi–input virtual temperature

As before, choose C_z resonant with this virtual qubit, $\epsilon_z = \epsilon_v$, and couple them by the swap [paper, after Eq. (32)]

$$H_{\text{int}} = g(|0\rangle\langle 1|_v \otimes |1\rangle\langle 0|_{C_z} + \text{h.c.}), \quad (72)$$

which is the direct generalization of (39). The excited–state population of the virtual qubit is the product of the per–qubit occupations [paper Eq. (33)]

$$g_v = g_0(h_0) g_1(h_1) \cdots g_n(h_n), \quad g_i(0) = (1 + e^{-\beta_i \epsilon_i})^{-1}, \quad g_i(1) = 1 - g_i(0), \quad (73)$$

and $e^{-\beta_v \epsilon_v} = g_v / (1 - g_v)$ gives, after taking the log of the product [paper Eq. (34)],

$$\boxed{\beta_v = \frac{1}{\epsilon_z} \sum_{i=0}^n (-1)^{h_i} \beta_i \epsilon_i.} \quad (74)$$

Again a *linear combination of the input inverse temperatures*, with weights $(-1)^{h_i}\epsilon_i/\epsilon_z$ set by the gaps and the vector h . The steady-state output is the same nonlinear map as in Part III, [paper Eqs. (35)–(36)]

$$\beta_z^\infty = \frac{1}{\epsilon_z} \ln[Q(\beta_v)^{-1} - 1] = f(\beta_v) + O(\epsilon_z), \quad (75)$$

with Q from (56). The whole machine still computes “weighted sum, then sigmoid.”

15 Exact correspondence with the perceptron

A perceptron with inputs $x = (x_0, \dots, x_n)$ (with $x_0 = 1$ by convention) and weights $w = (w_0, \dots, w_n)$ outputs [paper Eq. (37)]

$$z = f(y), \quad y = \sum_{i=0}^n w_i x_i. \quad (76)$$

The two-step structure matches the neuron line for line:

Perceptron	Thermodynamic neuron
inputs x_i	input inverse temperatures β_i
weighted sum $y = \sum_i w_i x_i$	virtual temperature β_v , Eq. (74)
bias w_0	reference term $(-1)^{h_0}\beta_0\epsilon_0/\epsilon_z$
weights w_i	$(-1)^{h_i}\epsilon_i/\epsilon_z$
activation f (sigmoid)	modulator map, small- ϵ_z Eq. (60)
output z	output temperature β_z^∞

Concretely, with an overall energy scale α (defined below), the pre-activation is [paper Eq. (44)]

$$\beta_z^\infty = f(x) + O(\epsilon_z), \quad x = \alpha \left(w_0 + \sum_{i=1}^n w_i \beta_i \right), \quad (77)$$

exactly the sigmoid perceptron. Hence a single thermodynamic neuron implements precisely the linearly-separable functions, and (by choosing other thermalization models) a strict superset.

16 Designing a neuron: the algorithm

Because the neuron *is* a perceptron, we design it by training a perceptron and reading off the hardware. Given a target n -input Boolean function $R(x)$ [paper, Algorithm 1]:

1. Build the training set $D = \{(x^{(i)}, R(x^{(i)}))\}_{i=1}^{2^n}$.
2. Train a linear classifier (sigmoid perceptron) to obtain weights $w = (w_0, \dots, w_n)$.
3. Set the interaction vector from the signs of the weights, $h_k = 0$ if $w_k \geq 0$, else $h_k = 1$ [paper Eq. (39)].
4. Set the qubit energies [paper Eq. (40)]

$$\epsilon_k = \begin{cases} \alpha |\epsilon_z - \sum_{k=1}^n w_k| & k = 0, \\ \alpha |w_k| & \text{otherwise.} \end{cases} \quad (78)$$

5. Set the reference (bias) temperature [paper Eq. (41)]

$$\beta_0 = \frac{|w_0|}{|\epsilon_z - \sum_{k=1}^n w_k|}. \quad (79)$$

Substituting into (74) reproduces $\beta_v = \frac{\alpha}{\epsilon_z}(w_0 + \sum_k w_k \beta_k)$ [paper Eqs. (42)–(43)], i.e. the trained hyperplane. The scale $\alpha > 0$ is the “quality knob”: it rescales all gaps, sharpening the threshold (better classification) at the price of proportionally more dissipation — the same accuracy/dissipation trade-off of Sec. 13, now global. For the NOT gate one finds $\alpha = \epsilon_1$.

How to implement — design + simulate a neuron

Train any logistic/perceptron classifier on the truth table (e.g. `sklearn`); take $\mathbf{w}$; map to $\mathbf{h}$, `eps_k`, `beta0` by the boxed formulas; build $H_0 + H_{\text{int}}$ (with the generalized swap (72)) plus reset baths in QuTiP; integrate the two-tier dynamics of Sec. 9; read β_z^∞ and decode with (63). Sweeping the inputs reproduces the response surfaces of [paper Figs. 6–7].

17 Worked examples

Example 17.1 (NOR, $n = 2$, paper Eqs. (45)–(46)). The separating hyperplane $x_1 + x_2 = \frac{1}{2}$ gives weights $w = (1, -2, -2)$. Then

$$h = (0, 1, 1), \quad \epsilon = \alpha(\epsilon_z + 4, 2, 2), \quad \beta_0 = (\epsilon_z + 4)^{-1}, \quad (80)$$

and the virtual temperature becomes $\beta_v = \alpha(1 - 2\beta_1 - 2\beta_2)$. The response $\beta_z^\infty(\beta_1, \beta_2)$ matches the NOR truth table. Because NOR is functionally complete, networks of these neurons are universal.

Example 17.2 (3-MAJORITY, $n = 3$, paper Eqs. (47)–(48)). Training gives $w = (-4, 3, 3, 3)$, hence

$$h = (1, 0, 0, 0), \quad \epsilon = \alpha(\epsilon_z + 12, 3, 3, 3), \quad \beta_0 = (\epsilon_z + 12)^{-1}, \quad (81)$$

and $\beta_v = \alpha(4 - 3\beta_1 - 3\beta_2 - 3\beta_3)$. The response reproduces the majority function.

18 Networks: beyond linear separability (XOR)

A single neuron can only realize linearly-separable functions (the perceptron limitation). XOR is not separable, so it needs a *network*: write $\text{XOR} = \text{AND}(\text{NAND}, \text{OR})$ and wire three neurons accordingly [paper, Fig. 8].

The wiring problem and its fix. Feeding one neuron’s *finite* output bath B_z into the next neuron’s input would let unwanted (possibly back-flowing) currents corrupt the operation, and Eq. (56) would no longer hold. Two remedies:

- *Measurement gateway*: an external agent reads β_z of the first neuron and couples the second neuron’s input qubit to an *infinite* bath at β_{hot} or β_{cold} accordingly. Simple, but sacrifices autonomy.
- *Clock gateway* [paper, Supp. C]: a stopwatch alternately opens/closes the coupling between B_z and the next input qubit. Define t_{steady} (time for neuron 1 to reach β_z^∞ with the link off) and t_{const} (interval over which B_z stays effectively constant when used as neuron 2’s input). Turning the link on for t_{const} after each t_{steady} makes B_z behave as an infinite bath to neuron 2, with no spurious back-currents. Realized with an autonomous thermal clock, the network stays fully autonomous.

This clock-gated readout is exactly the mechanism the Floquet-buffer extension internalizes: the J_0 -zero gating of the output buffer plays the role of $t_{\text{const}}/t_{\text{steady}}$, replacing the external stopwatch by the drive itself.

End of Part IV. The n -input neuron computes $\beta_z^\infty \approx f(\alpha(w_0 + \sum_i w_i \beta_i))$, a sigmoid perceptron; its hardware (h, ϵ_k, β_0) is read off a trained classifier; networks (clock-gated) reach any Boolean function. This is the complete autonomous baseline; the Floquet buffer aims to improve its accuracy/dissipation frontier and to make composition native.

Appendices

A Notation and symbol table

Symbol	Meaning	First appears
ϵ_i	energy gap of qubit C_i	(4)
β_i	inverse temperature of bath B_i	(5)
$g(\beta\epsilon)$	Fermi-Dirac / sigmoid occupation	(7)
$\tau(\beta)$	Gibbs state	(5)
$\mathcal{L}^{(k)}$	reset dissipator for bath k	(19)
γ_k, μ, μ'	thermalization rates	(19), (40)
β_v, ϵ_v	virtual temperature and gap	(30)–(32)
χ	collector coupling strength	(36)
j_k	heat current into bath k	(20)
$\dot{\Sigma}$	entropy production rate	(26)

B List of approximations

Tag	Name	Regime of validity
(A1)	Born / weak coupling	$\gamma \ll \epsilon$, factorized state
(A2)	Markov / short memory	bath correlation time $\tau_c \ll 1/\gamma$
(A3)	secular / RWA	$\gamma \ll \epsilon$ (resolved gap)
(A4)	reset thermalization	steady-state populations only
(A5)	energy-preserving H_{int}	$\chi \ll \epsilon$
(A6)	local master equation	$\chi \ll \epsilon$ (else global)

C Full Born–Markov–secular derivation

We fill in the steps compressed in Sec. 2.3. Start from (13) and trace over the bath. Writing $H_{SB} = \sum_\alpha A_\alpha \otimes B_\alpha$ (here $A = \sigma_x$, one term), the second-order generator is

$$\dot{\tilde{\rho}}(t) = - \int_0^t ds \text{Tr}_B [\tilde{H}_{SB}(t), [\tilde{H}_{SB}(s), \tilde{\rho}(s) \otimes \rho_B]]. \quad (82)$$

Expanding the double commutator and using $\text{Tr}_B [B_\alpha \rho_B] = 0$ leaves products of the *bath correlation functions*

$$C_{\alpha\beta}(t-s) = \text{Tr}_B [\tilde{B}_\alpha(t) \tilde{B}_\beta(s) \rho_B], \quad (83)$$

which depend only on $\tau = t - s$ because ρ_B is stationary. Decompose the system coupling operator into eigenoperators of H_S ,

$$A(\omega) = \sum_{\varepsilon' - \varepsilon = \omega} \Pi(\varepsilon) A \Pi(\varepsilon'), \quad \tilde{A}(\omega, t) = e^{-i\omega t} A(\omega), \quad (84)$$

with Π the projectors onto the eigenspaces of H_S ; for the qubit $A(\pm\epsilon) = \sigma_{\mp}$. Making the Markov replacement (A2) $\tilde{\rho}(s) \rightarrow \tilde{\rho}(t)$ and extending the integral to infinity, one gets the Redfield equation with the half-Fourier rates

$$\Gamma(\omega) = \int_0^\infty d\tau e^{i\omega\tau} C(\tau) = \frac{1}{2}\gamma(\omega) + iS(\omega), \quad (85)$$

where $\gamma(\omega)$ is the full Fourier transform (14) (a real, positive relaxation rate) and $S(\omega)$ its imaginary part (the Lamb shift). The secular approximation (A3) drops all cross terms with $\omega \neq \omega'$ (they rotate at $e^{i(\omega' - \omega)t}$), yielding the diagonal, completely positive GKSL form

$$\dot{\rho} = -i[H_S + H_{LS}, \rho] + \sum_{\omega} \gamma(\omega) \left(A(\omega) \rho A^\dagger(\omega) - \frac{1}{2} \{ A^\dagger(\omega) A(\omega), \rho \} \right), \quad (86)$$

with $H_{LS} = \sum_{\omega} S(\omega) A^\dagger(\omega) A(\omega)$. For the qubit this is exactly (15) with $\gamma_{\downarrow} = \gamma(\epsilon)$, $\gamma_{\uparrow} = \gamma(-\epsilon)$, and the KMS relation $\gamma(-\omega) = e^{-\beta\omega} \gamma(\omega)$ gives (16). The Lamb shift $H_{LS} \propto \sigma_+ \sigma_-$ merely renormalizes ϵ and is usually absorbed into it. See Ref. [5], Ch. 3, for the full treatment.

D The reset model as a collisional process

The reset dissipator (19) arises from a repeated-interaction (collisional) model. In each interval dt , qubit k either (prob. $\gamma_k dt$) is swapped for a fresh bath qubit in state $\tau(\beta_k)$, or (prob. $1 - \gamma_k dt$) is left untouched:

$$\rho(t + dt) = (1 - \gamma_k dt) \rho(t) + \gamma_k dt (\text{Tr}_k[\rho(t)] \otimes \tau(\beta_k)). \quad (87)$$

Rearranging and taking $dt \rightarrow 0$ gives exactly $\dot{\rho} = \gamma_k (\text{Tr}_k[\rho] \otimes \tau(\beta_k) - \rho)$. The map is manifestly completely positive and trace preserving, and has $\tau(\beta_k)$ as its unique fixed point on qubit k , so it satisfies local detailed balance (25) by construction. This is why the reset model can stand in for the microscopically derived dissipator whenever only steady-state populations and their thermodynamics matter (approximation (A4)).

E Numerical implementation in QuTiP

A minimal, self-contained recipe that reproduces the NOT gate.

Operators. With ordering $(C_0 C_1 C_z)$:

```
import numpy as np, qutip as qt
I2 = qt.qeye(2); sm = qt.destroy(2); sp = sm.dag(); n = sp*sm
def op(o,site): L=[I2,I2,I2]; L[site]=o; return qt.tensor(L)
n0,n1,nz = op(n,0),op(n,1),op(n,2)
H0 = eps0*n0 + eps1*n1 + eps_z*nz # free Hamiltonian, Eq.(A? / HS)
k100 = qt.tensor(qt.basis(2,1),qt.basis(2,0),qt.basis(2,0))
k011 = qt.tensor(qt.basis(2,0),qt.basis(2,1),qt.basis(2,1))
Hint = chi*(k100*k011.dag() + k011*k100.dag()) # resonant swap, Eq.(Hint)
H = H0 + Hint
assert (qt.commutator(H0,Hint)).norm() < 1e-12 # resonance sanity check
```

Reset dissipators. Build collapse operators reproducing the reset fixed point for each bath (rate γ_i , occupation $g_i = g(\beta_i \epsilon_i)$):

```
def bath(site, gamma, g):      # detailed-balanced up/down channels
    return [np.sqrt(gamma*(1-g))*op(sm,site),  # emission (down)
            np.sqrt(gamma*g) *op(sp,site)]     # absorption(up)
g0 = 1/(1+np.exp(beta0*eps0)); g1 = 1/(1+np.exp(beta1*eps1))
gz = 1/(1+np.exp(betaz*eps_z))
c_ops = bath(0,gamma,g0) + bath(1,gamma,g1) + bath(2,mu,gz)
rho_ss = qt.steadystate(H, c_ops)                # collector-only NESS
p_z = qt.expect(nz, rho_ss)                      # excited pop of C_z
beta_z = np.log((1-p_z)/p_z)/eps_z               # -> approaches beta_v
```

Two-tier dynamics (the full gate). Because $\gamma \gg \mu, \mu'$ (Sec. 9), do not integrate the stiff full model directly. Instead, for a grid of β_z values compute the fast steady-state currents $j_C(\beta_z) = \mu \epsilon_z [g_z(\beta_z) - g_z(\beta_v)]$ and $j_M(\beta_z) = \mu' \epsilon_z [g_z(\beta_z) - g_z(\beta_r)]$, then integrate the scalar slow ODE (48) $\dot{\beta}_z = -(\beta_z^2/C)(j_C + j_M)$ with a standard (non-stiff) solver; its fixed point is (56). Sweeping $\beta_1 \in \{\beta_{\text{hot}}, \beta_{\text{cold}}\}$ and decoding with (63) yields the truth table; sweeping β_1 continuously yields the transfer curve [paper Fig. 3B].

Currents, entropy production, error. Heat currents follow from (20), $j_k = \text{Tr}[H \mathcal{L}^{(k)} \rho]$; entropy production from (26), $\dot{\Sigma} = -\sum_k \beta_k j_k$ in the NESS. The readout error (66) is obtained by convolving the transfer curve with the Gaussian $\mathcal{N}(\beta_z^\infty, C)$ and integrating against the guard bands. Plotting $\langle \Sigma \rangle$ vs $\langle \xi \rangle$ over a sweep of ϵ_1 reproduces the trade-off of [paper Fig. 3C].

How to implement — validation checklist

(i) $\| [H_0, H_{\text{int}}] \| = 0$ (resonance). (ii) Collector-only steady state gives $\beta_z \rightarrow \beta_v$ within tolerance. (iii) First law per bath: $\sum_k j_k = 0$ in the NESS (energy conservation). (iv) $\dot{\Sigma} \geq 0$ at every input. (v) Transfer curve from the ODE matches the closed form (56). Only after all five pass should any physics be trusted.

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